

Spring REST using Spring Boot 3 | Create authentication service that returns JWT

Cognizant DN 5.0

The image displays two screenshots of an IDE (Visual Studio Code) showing the development of a Spring REST application. The top screenshot shows the `AuthenticationController.java` file, and the bottom screenshot shows the `CountryController.java` file.

Top Screenshot: `AuthenticationController.java`

```
src > main > java > com > cognizant > spring_rest_handson > controller > AuthenticationController.java > Language Support for Java(TM) by Red Hat > AuthenticationController > authenticate()
1 package com.cognizant.spring_rest_handson.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 import com.cognizant.spring_rest_handson.model.AuthenticationResponse;
7 import com.cognizant.spring_rest_handson.util.JwtUtil;
8
9 @RestController
10 public class AuthenticationController {
11
12     @GetMapping("/authenticate")
13     public AuthenticationResponse authenticate() {
14
15         String token = JwtUtil.generateToken(username = "user");
16
17         return new AuthenticationResponse(token);
18     }
19 }
```

Bottom Screenshot: `CountryController.java`

```
src > main > java > com > cognizant > spring_rest_handson > controller > CountryController.java > Language Support for Java(TM) by Red Hat > CountryController > getCountryIndia()
13 import com.cognizant.spring_rest_handson.model.Country;
14 import com.cognizant.spring_rest_handson.service.CountryService;
15
16 @RestController
17 public class CountryController {
18
19     private static final Logger LOGGER =
20         LoggerFactory.getLogger(CountryController.class);
21
22     private final CountryService countryService;
23
24     public CountryController(CountryService countryService) {
25         this.countryService = countryService;
26     }
27
28     @GetMapping("/country")
29     public Country getCountryIndia() {
30
31         LOGGER.info("Start");
32
33         Country country = new Country(code: "IN", name: "India");
34
35         LOGGER.info("End");
36
37         return country;
38     }
39
40     @GetMapping("/countries")
41     public List<Country> getAllCountries() {
42
43         LOGGER.info("Start");
44
45         List<Country> countries = new ArrayList<>();
46
47         countries.add(new Country(code: "IN", name: "India"));
48         countries.add(new Country(code: "US", name: "United States"));
49         countries.add(new Country(code: "JP", name: "Japan"));
50         countries.add(new Country(code: "DE", name: "Germany"));
51     }
52 }
```

File Edit Selection View Go Run Terminal Help

spring-rest-handson

try.java AuthenticationResponse.java JwtUtil.java AuthenticationController.java pom.xml SecurityConfig.java CountryController.java 1

SPRING-REST-HANDSON

src

main

java

com

cognizant

spring_rest_hands...

model

AuthenticationRespo...

Country.java

security

SecurityConfig.java

service

CountryService.java

util

JwtUtil.java

SpringLearnApplicatio...

resources

static

templates

application.properties

test

target

.gitattributes

.gitignore

HELP.md

mvnw

mvnw.cmd

pom.xml

OUTLINE

TIMELINE

LOGICAL STRUCTURE

JAVA PROJECTS

MAVEN

try.java

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```
<?xml version="1.0" encoding="UTF-8" standalone="no" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.cognizant</groupId>
    <artifactId>spring-rest-handson</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <packaging>war</packaging>
    <name>spring-rest-handson</name>
    <description>Spring REST Handson</description>
    <properties>
        <java.version>17</java.version>
    </properties>
    <dependencies>
        <!-- Spring Boot Starters -->
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter</artifactId>
            <version>2.7.0</version>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
            <version>2.7.0</version>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-security</artifactId>
            <version>2.7.0</version>
        </dependency>
        <dependency>
            <groupId>io.jsonwebtoken</groupId>
            <artifactId>jjwt-api</artifactId>
            <version>0.12.6</version>
        </dependency>
        <dependency>
            <groupId>io.jsonwebtoken</groupId>
            <artifactId>jjwt-impl</artifactId>
            <version>0.12.6</version>
        </dependency>
        <dependency>
            <groupId>io.jsonwebtoken</groupId>
            <artifactId>jjwt-jackson</artifactId>
            <version>0.12.6</version>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-test</artifactId>
            <version>2.7.0</version>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>
</project>
```

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```
package com.cognizant.spring_rest_handson.util;

import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.security.Keys;

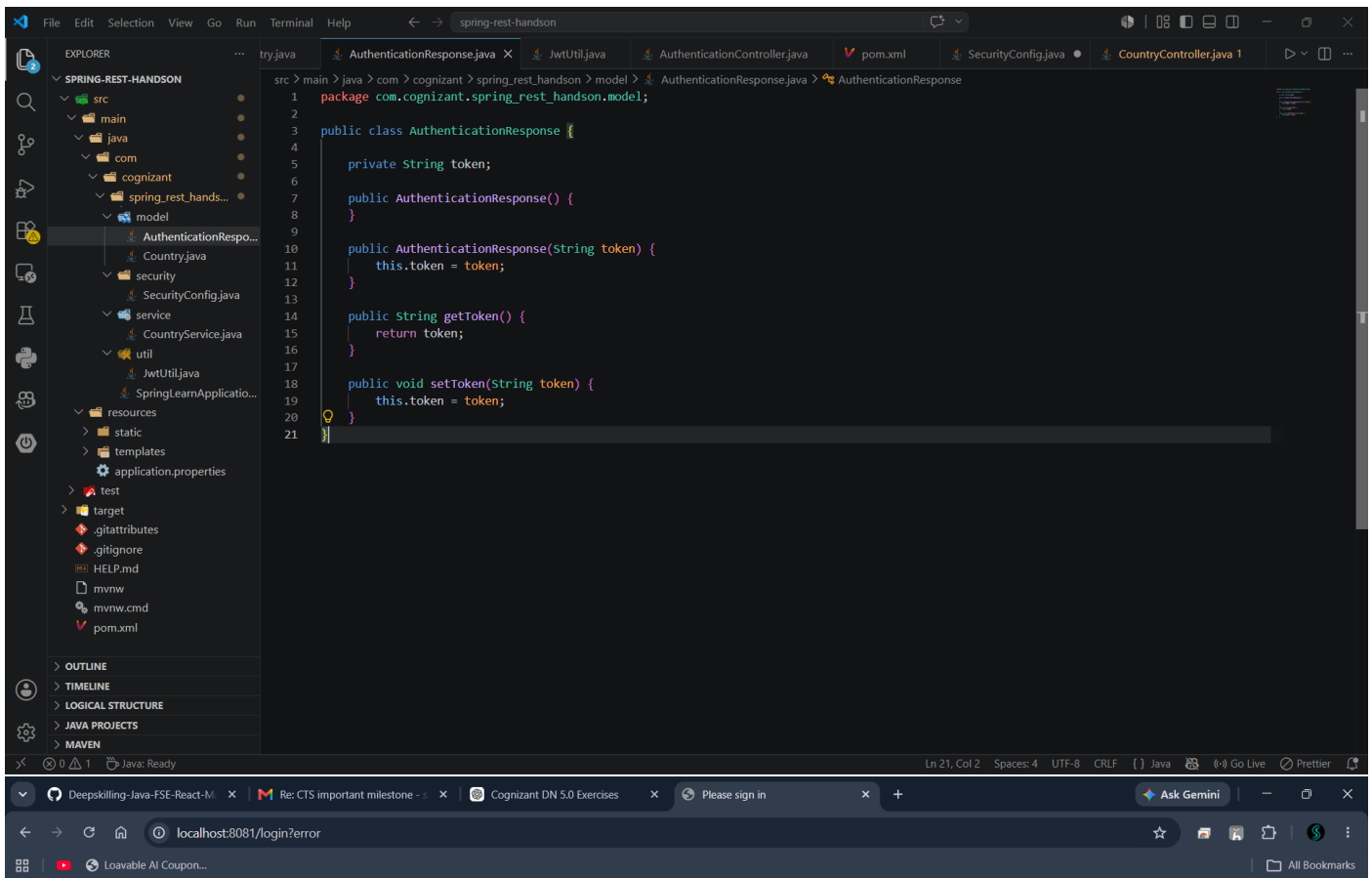
import javax.crypto.SecretKey;
import java.util.Date;

public class JwtUtil {

    private static final SecretKey KEY =
        Keys.hmacShaKeyFor(
            "ThisIsMySecretKeyForJwtAuthentication123456".getBytes()
        );

    public static String generateToken(String username) {
        return Jwts.builder()
            .subject(username)
            .issuedAt(new Date())
            .expiration(new Date(System.currentTimeMillis() + 600000))
            .signWith(KEY)
            .compact();
    }
}
```

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Password

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